Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/19/2007
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Project Title:	Checkpoint Reduced Cell Sensitivity to High Energy Particles-Induced Killing		
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Zip Code:	30322	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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Contact Monitor:		**Contact Phone:**	
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Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: End date changed to 6/30/07 per A. Chu (5/06)		
Key Personnel Changes/Previous PI:			
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Task Description:	This application is in response to the NASA Program NRA 02-OBPR-02 and addresses research related to both Bioastronautics and Fundamental Biology. Radiation-induced DNA double strand breaks (DSBs) are a severe threat to genomic integrity that determines radiosensitivity (short term effect) and carcinogenesis (long term effect). Two major DNA DSB repair pathways exist in eukaryotic cells, homologous recombination repair (HRR) and non-homologous end joining (NHEJ). After induction of DSBs, proliferating cells actively slow down cell cycle progression via checkpoint activation to provide time for repair. However, it is not known whether checkpoint activation benefits NHEJ and HRR equally, or whether it is specifically adapted to assist just one of these processes, therefore reducing DSBs-induced genomic instability. The long-term objective of the present proposal is to elucidate the molecular mechanisms by which checkpoint activation facilitates DNA repair and therefore reduces the high-energy (HZE) particles-induced genomic integrity in mammalian cells. In this proposal, the central hypothesis is that CHK1-regulated checkpoints, in an ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. The preliminary results from sparsely ionizing radiation provide evidence to support this hypothesis and form the foundation for the experimental design. Three Specific Aims are pursued: Specific aim 1: Determine whether ATR, the upstream kinase of CHK1, affects HZE particles-induced killing by facilitating HRR. We will examine HRR (by using the GFP-I-SceI restriction enzyme system), NHEJ (by using asymmetric field inversion gel electrophoresis, AFIGE), CHK1 activity (by using kinase assays) and the effect of CHK1 siRNA on HZE particles-induced killing (by using clonogenic assay) in ATR kinase dead (ATR-kd) human fibroblasts. Specific aim 2: Determine whether the protective effect of the CHK1-regulated checkpoint on HZE particles-induced killing is mediated by facilitating HRR. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to A1-5 cells, a rat embryo fibroblast line in which CHK1 is highly expressed. We will examine HRR, NHEJ, CHK1 activity and HZE particles-induced killing in CHK1-kd cell lines. Specific aim 3: Determine whether the effect of CHK1 on HZE particles-induced killing is ATM independent. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to AT5BIVA (ATM-/-) human fibroblast cells. We will test HZE particles-induced killing, HRR, NHEJ, CHK1 activity and checkpoint activation in the CHK1-kd AT cell lines following HZE particles radiation. This approach will clarify the relationships among ATR/CHK1-regulated checkpoint activation, NHEJ and HRR. Such information will provide means for protecting mammals from HZE particles-induced killing in a short term and from HZE particles-induced carcinogenesis in a long term by activating a particular pathway (ATM or ATR-regulated checkpoints, NHEJ or HRR) depending on the genetic background of the mammals. The proposal addresses a basic research topic related to the protection of individuals from the hazards of ionizing radiation in outer space. The proposed basic research is within the framework defined by the Space Radiation Health Program, and attempts to establish the scientific basis for radiation protection of humans engaged in the exploration of outer space. It is a ground-based program that studies the hypothesis that CHK1-regulated checkpoints, in an ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. We believe that our results from this proposal will clarify the relationships among ATR/CHK1-regulated checkpoint activation, NHEJ and HRR. Once we understand the relationships among checkpoint activation, HRR and NHEJ, we will be able to protect cells from HZE particles-induced killing by activating the particular pathway (ATM- or ATR/CHK-regulated checkpoint, HRR or NHEJ) of the cells with different genetic backgrounds. We expect that these results will have the potential for reducing the HZE particles-induced mutagenesis and carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results from these experiments will clarify the role of the ATR/CHK1 pathway, the most important checkpoint proteins in mammalian cells, in protecting cells from ionizing radiation (IR)-induced killing.
Task Progress:	This project has been funded for three years (02-OBPR-02-0000-0037). We have followed the proposal to carry out some experiments as described in our specific aims. Radiation-induced DNA double strand breaks (DSBs) are a severe threat to genomic integrity that determines radiosensitivity (short-term effect) and carcinogenesis (long-term effect). Two major DNA DSB repair pathways exist in eukaryotic cells, homologous recombination repair (HRR) and non-homologous end joining (NHEJ). After induction of DSBs, proliferating cells actively slow down cell cycle progression via checkpoint activation to provide time for repair. However, it is not known whether checkpoint activation benefits NHEJ and HRR equally, or whether it is specifically adapted to assist just one of these processes therefore reducing DSBs-induced genomic instability. Till now, we have successfully examined the effect of ATR or CHK1 on HRR (by using the GFP-I-SceI restriction enzyme system) and NHEJ (by using asymmetric field inversion gel electrophoresis, AFIGE). We have also examined the effect of ATR or CHK1 on HZE particles-induced killing. The results show that the doxycycline (dox)-induced ATR kinase dead (ATRkd) cells have the similar inductions and rejoining rates of DNA DSBs compared with the cells without dox-induction, although the dox-induced ATRkd cells are more sensitive to IR and have the deficient S as well as G2 checkpoints. We also show here that the dox-induced ATRkd cells have a lower HRR efficiency compared with the cells without tet-induction. These results indicate that the effects of ATR on cell radiosensitivity is independent of NHEJ but is linked to HRR that might be affected by the deficient S and G2 checkpoints. Also, by using CHK1 kinase dead human cell lines established in our laboratory, we show that although these human cell lines have different CHK1 activities with different sensitivities to IR-induced killing and G2 accumulation, all these cell lines show similar inductions and rejoining rates of DNA DSBs. These results indicate that the different radiosensitivities and G2 checkpoint responses in these cell lines are independent of NHEJ. In addition, by combining I-SceI-induced-DNA DSB system and siRNA approach, we also show that knocking down Chk1 decreases the efficiency of HRR, associated with the sensitivity to IR-induced killing. Taken together, the results demonstrate that the role of the ATR/CHK1 pathway in affecting the sensitivity of cells to IR-induced killing is linked to HRR. DNA DSBs induced by either ionizing radiation (IR) or HZE particles are the most severe threat for genomic integrity. Unrepaired DSBs lead to broken chromosomes and/or gross chromosomal rearrangements, which are correlated with radiosensitivity, and are associated with tumor formation in mammals. Experimental evidence supports the causal link between DSBs and the induction of gene mutations, chromosomal aberrations and cell transformation. Two major DSB repair pathways exist in eukaryotes, HRR and NHEJ. These mechanisms are conserved in evolution, but the relative contribution depends on the organism species. DNA DSBs in yeast are primarily repaired via HRR. Originally, NHEJ was considered to be the only major mechanism for repairing DNA DSBs in mammals. Only recently has it become evident that HRR also has an important role in repairing DNA DSBs in mammals. The mechanisms underlying these two repair pathways, as well as their relationships and interactions during DSB removal, are not yet completely understood.

Bibliography Type:	Description: (Last Updated: 07/07/2021)
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